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Dissociative Adsorption of H_2 on PtRu Bimetallic Surfaces

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Abstract

We present a theoretical study of the dissociative adsorption of molecular hydrogen on PtRu bimetallic surfaces based on Density Functional Theory (DFT) calculations. We focus on the reactivity of a pseudomorphic Pt monolayer deposited on Ru(0001), Pt_{1ML}/Ru(0001), for which we have obtained a minimum activation energy barrier for H_2 dissociation E_b=0.32 eV, i.e. $\sim$ 0.3 eV and 0.26 eV higher than on Ru(0001), and Pt(111) respectively. Accordingly, the initial sticking probability for low energy impinging molecules ($E_i \lesssim 0.1$ eV) derived from classical trajectory calculations is various orders of magnitude smaller than on Ru(0001) in apparent contradiction with available experimental data. However, undercoordinated Pt atoms in the borders of Pt pseudomorphic islands and isolated Ru atoms or small Ru aggregates in a Pt-rich two-dimensional Pt_xRu_{1-x} surface alloy provide active sites allowing non-activated H_2 dissociative adsorption. These two possible defects in a full pseudomorphic Pt monolayer deposited on Ru(0001) might be the responsible for the relatively high initial sticking probabilities obtained experimentally for $\sim$ 1.0-1.2 ML Pt evaporated over Ru(0001), only $\sim$ 30-65 % smaller than on Ru(0001).

1 Introduction

Bimetallic surfaces are attractive in catalysis because by varying the constituents, stoichiometry, and preparation conditions, it is possible to produce a huge variety of materials with specific properties.[1–5] Pt-based bimetallic surfaces are of particular interest because they are largely employed in many catalytic and electrocatalytic applications (see 6 and references therein). An example is the oxygen reduction (OR) reaction, of great importance for applications in fuel cells,[7] for which Pt was considered for a long time the most active catalyst.[8] However, it has been recently shown that alloying Pt with e.g. Ni, Co, Cu, or Ru, it is possible to achieve a higher OR activity than with pure Pt. Under reaction conditions, Pt-based catalysts often consist in supported core-shell particles with a Pt shell and a bimetallic core.

Thus, studies on the reactivity of pseudomorphic Pt overlayers on the surface of another metal, M, and monolayer Pt_xM_{1-x}/M surface alloys ($x \leq 1$) are of great importance in the search of, and for a fundamental understanding of the activity of bimetallic catalysts.

Behm and co-workers have performed a detailed and systematic experimental study of the structure and reactivity of bimetallic PtRu surfaces resulting under various preparation conditions (see e.g. refs. 9,10). On the one hand, they have shown that Pt evaporation at room temperature followed by annealing at $\sim$ 650 K gives rise to compact pseudomorphic Pt islands of variable thickness.[11] In spite of the preferentially pseudomorphic growth of Pt on Ru(0001) up to at least four monolayers (ML),[12] for instance, for 1.2 ML of Pt, only 66% of the surface is covered by a single Pt monolayer, whereas 25% corresponds to two-layer Pt islands, and 9% to Ru atoms exposed due to vacancies in the Pt-monolayer areas.[13] Accordingly, such a bimetallic surface exposes Pt patches of variable thickness (from zero to two Pt layers) and reactivity,[14] as well as undercoordinated Pt atoms (in the borders of all these regions) which are likely more reactive than those in flat terraces.[15] On the other hand, deposition of Pt up to 1 ML coverage at room temperature followed by annealing at $\sim$ 1300 K leads to the formation of a Pt_xRu_{1-x} alloy characterized by a statistical distribution of Pt and Ru atoms in the outermost surface layer.[9,10] This provokes a variety of Pt_n and Ru_m aggregates presenting different local reactivities depending on their composition (ensemble effect) and surroundings (ligand effects).[16] Thus, in both cases (pure Pt islands, and Pt_xRu_{1-x} monolayer alloys), predicting and/or understanding the origin of the global reactivity of the bimetallic surface is not straightforward in spite of the precise morphology characterization achieved through scanning tunnelling microscopy (STM) experiments.[9,10] Particularly interesting is the fact that the combinative desorption peak of D_2 from 1.0-1.2 ML Pt deposited on Ru(0001) obtained in temperature programmed desorption (TPD) experiments is $\sim$ 200 K smaller than from Ru(0001). This puts in evidence a binding energy of atomic hydrogen on the pseudomorphic Pt monolayer much smaller than on Ru(0001) that is however, accompanied by a reduction of the initial reactive sticking probability of D_2

of only 30 % - 65 % depending on the annealing temperature after Pt deposition.[17,18] This suggests a weak correlation between the binding energy of atomic hydrogen and the initial dissociative adsorption probability of molecular hydrogen on PtRu bimetallic surfaces which has been also observed in PdRu ones.[19]

In order to characterize the reactivity of various model PtRu bimetallic surfaces, Density Functional Theory (DFT) has been used to compute the binding energy of various adsorbates (including atomic hydrogen) on pseudomorphic Pt overlayers on Ru(0001),[11] and substitutional PtRu surfaces (see e.g. refs. 20,21). In general, it has been found that H, CO, OH, O, and O_2 bind to Pt_{1ML}/Ru(0001) more weakly than on Pt(111), Ru(0001) and Pt-multilayers deposited on Ru(0001).[11,12,14,22] This is explained in terms of strain, ligand, and ensemble effects.[1,3,16,23] For instance, pseudomorphic overlayers of Pt deposited on Ru(0001) are compressed with respect to Pt(111). This entails a larger overlap between the Pt d-orbitals and consequently, a broadening and down-shift of the d-band which according with the d-band model,[24] turns the pseudomorphic monolayers of Pt deposited on Ru(0001) less reactive than Pt(111).[25] In addition, the attractive Pt-Ru interaction stronger than the Pt-Pt one[14] (vertical ligand effect), makes the binding energies of adsorbates on a single monolayer of Pt, smaller than on Pt multilayers (a stronger bond with atoms of the underlying metal layers, tends to weaken the bond between the outermost-layer metal atoms and adsorbates). Since the bond of a hydrogen atom with Pt(111) is weaker than with Ru(0001), the arguments above explain why the binding energy on a single monolayer (1ML) of Pt deposited on Ru(0001), Pt_{1ML}/Ru(0001), is smaller (adsorption less stable) than on any other number of pseudomorphic Pt overlayers, and also smaller than on Pt(111) and Ru(0001).

Though the binding energy of H atoms (the products of H_2 dissociation) certainly provides valuable information about the reactivity of PtRu surfaces, its connection with the dissociative adsorption probability of H_2 molecules impinging the surface is not straightforward. In contrast, activation energy barriers are certainly more directly connected with surface reactivity, but still dissociative adsorption probabilities are sometimes hardly pre-

dictable without accounting for dynamical effects.[26,27] In this work, we present a DFT study of dissociative adsorption of H_2, as a probe of the reactivity of bimetallic PtRu surfaces. We investigate dissociation pathways, in order to estimate the activation energy barriers encountered by H_2 molecules on various possible structures of PtRu bimetallic surfaces. The study is focused on Pt_{1ML}/Ru(0001) for which we have also performed quasi-classical trajectory (QCT) calculations using a potential energy surface (PES) interpolated through the corrugation reducing procedure (CRP).[28] The origin of the choice of H_2/Pt_{1ML}/Ru(0001) for such an extended investigation (including dynamics) is two-fold. On the one hand, among the bimetallic surfaces made of n Pt overlayers deposited on Ru(0001), Pt_{1ML}/Ru(0001) is the one whose H adsorption energy differs most from those on Pt(111) and Ru(0001).[11] Accordingly, pronounced differences in the H_2 dissociation process with respect to the latter two pure metal surfaces for which dynamical studies already exist[29–31] are not unexpected. On the other hand, the reactivity of a full single monolayer of Pt on Ru(0001) is necessary for a better understanding of the properties of Pt-rich Pt_xRu_{1-x} monolayer alloys obtained for the annealing temperature of 1300 K,[17] as well as those of Pt-monolayer islands and terraces obtained for the annealing temperature of 650 K for up to $\sim$ 2 ML Pt evaporated on Ru(0001).[13] In addition to H_2/Pt_{1ML}/Ru(0001), here we investigate H_2 dissociation pathways on: i) Pt_{2ML}/Ru(0001), ii) underlying Ru atoms exposed due to the existence of Pt-vacancies in Pt overlayers, iii) undercoordinated Pt atoms (e.g. in the borders of Pt islands), and iv) isolated aggregates of one and two Ru atoms in a Pt-rich Pt_xRu_{1-x} overlayer. In the light of these results, we discuss on the origin of the experimental initial sticking probability of molecular hydrogen on the PtRu bimetallic surfaces for $\sim$ 1ML Pt coverage reported by Behm and co-workers.[17,18]

2 Computational details

DFT calculations were performed using the Vienna ab-initio simulation package (VASP),[32–35] using projected augmented-wave (PAW) pseudopotentials to describe the electron-ionic core interactions. Electronic exchange and correlation was described within the generalized gradient approximation (GGA) using the functional proposed by Perdew, Burke and Ernzerhof (PBE).[36,37] The obtained theoretical lattice constants for bulk Ru (hcp crystal) and Pt (fcc crystal) are $a_{\mathrm{Ru}} = 2.71$ Å, $c_{\mathrm{Ru}} = 4.28$ Å, and $a_{\mathrm{Pt}} = 3.97$ Å, which differ from the experimental values by less than 1.5 %.[38] Thus, the corresponding nearest neighbor (NN) distance between atoms in the (111) and (0001) planes of bulk Pt and Ru are $\Delta_{\mathrm{Pt(111)}}=a_{\mathrm{Pt}}/\sqrt{2}=2.81$ Å, and $\Delta_{\mathrm{Ru(0001)}}=a_{\mathrm{Ru}}=2.71$ Å respectively.

The PtRu bimetallic surfaces were always represented by a slab of five metal layers (e.g. 1Pt+4Ru for $\mathrm{Pt_{1ML}}$/Ru(0001) and 2Pt+3Ru for $\mathrm{Pt_{2ML}}$/Ru(0001)), characterized by the theoretical intra-layer NN distance between metal atoms equal to $\Delta_{\mathrm{Ru(0001)}}$.

In order to avoid artifacts due to the use of periodic boundary conditions in the direction perpendicular to the surface, we have taken a $\sim$ 14 Å width vacuum space between consecutive slabs. The calculations for Ru(0001), $\mathrm{Pt_{1ML}}$/Ru(0001), and $\mathrm{Pt_{2ML}}$/Ru(0001) were carried out using a (2 × 2) unit cell, whereas for $\mathrm{Pt_xRu_{1-x}}$/Ru(0001) (x=0.78, and x=0.89), and for the Pt overlayers with vacancies (see below) we have used a (3 × 3) unit cell. In calculations for 2×2, and 3×3 unit cells we have used 9×9×1 and 7×7×1 meshes of k-points respectively, selected according to the Monkhorst and Pack method.[39] We have used an energy cut-off equal to 400 eV, and an electronic smearing $\sigma = 0.2$ eV, following the Methfessel an Paxton scheme.[40] In all the calculations allowing surface relaxation, we have optimized the positions of the atoms in the three topmost layers of the slab, whereas the distance between the two bottom layers was kept fixed in the value corresponding to bulk Ru.

Due to the lattice-parameter mismatch between Ru and Pt, pseudomorphic overlayers of Pt on Ru(0001) are compressed by $\sim$ 3.5 % with respect to Pt(111). This strain entails a

broadening and down-shift of the d-band projected on the outermost layer surface atoms. For instance, for pure Pt(111) the d-band center is located at -2.23 eV (i.e. 2.23 eV below the Fermi level), whereas in the case of a Pt(111) surface characterized by a NN distance equal to the value corresponding to Ru(0001) (hereafter referred to as Pt(111)@Ru), it is located at -2.37 eV. In addition, in the case of Pt_{1ML}/Ru(0001) the center of the projected d-band is located at -2.61 eV, i.e. even lower than for Pt(111)@Ru which means that both ligand and strain effects produce a down-shift of the Pt-projected d-band. Interestingly, almost the same position of the d-band center is found for two and more pseudomorphic Pt overlayers on Ru(0001) which indicates that for more than a single Pt overlayer, ligand effects are small, i.e. topmost-layer Pt atoms interact weakly with the underlying Ru support. The dependence of the d-band center position for various Pt overlayers obtained with our DFT settings is the same as that of ref. 14.

As mentioned in Section 1, DFT total energies for H_2/Pt_{1ML}/Ru(0001) have been interpolated using the CRP in order to obtain a continuous representation of the molecule-surface PES. Within the rigid surface approximation, the H_2/Pt_{1ML}/Ru(0001) PES is six-dimensional (6D) and as usual, we describe it as a function of the cartesian coordinates of the molecular center of mass (X_{cm}, Y_{cm}, Z_{cm}), the H-H distance (r), the polar angle (θ), and the azimuthal angle (ϕ) of the H-H internuclear vector. We have interpolated a set of 322 DFT H/surface total energies corresponding to the surface sites indicated in Figure 1, and 3978 DFT H_2/surface total energies for 18 molecular configurations characterized by the following positions of the molecular center of mass and orientations:

i) on *top* with θ=0°, (θ,ϕ)=(90°,0°), and (90°,30°);

ii) on *bridge* with θ=0°, (θ,ϕ)=(90°,0°), (90°,60°), and (90°,90°);

iii) on *hollow* with θ=0°, (θ,ϕ)=(45°,30°), (45°,90°), (90°,0°), and (90°,30°);

iv) on *top-to-hollow* with θ=0°, (θ,ϕ)=(45°,30°), (45°,120°), (45°,210°), (90°,30°), and (90°,120°).

The accuracy of the interpolated PES was checked by comparison with DFT total energies for molecular configurations not included in the interpolation database. We have found differences $\lesssim 25$ meV for all the explored configurations which are energetically accessible for molecules with the low impact energies of interest in this work (i.e. $\lesssim 1$ eV). Then, the CRP PES was employed in QCT calculations, in order to evaluate the initial dissociative adsorption probability of H_2 on Pt_{1ML}/Ru(0001) as a function of the initial translational kinetic energy, E_i, and the angle of incidence, θ_i, and for various initial ro-vibrational molecular states $H_2(\nu,J)$. Dissociation was considered to take place whenever the internuclear distance reached the value r_{diss}= 2.3 Å with $dr/dt > 0$, and trajectories were considered as reflected when they reached the initial height above the surface (i.e. $Z_{\rm cm} = 6.0$ Å) with a velocity of the center of mass pointing to the vacuum. The probabilities were computed as the ratio of the number of trajectories leading to the analyzed reaction channel and the total number of computed trajectories (5000 per each initial condition).

3 Results and discussions

3.1 Pt_{1ML}/Ru(0001)

The interaction potential of atomic hydrogen with Pt_{1ML}/Ru(0001) as a function of the H position in front of the surface, characterized by the cartesian coordinates x, y, z, $V_{\rm H}(x, y, z)$, was computed using the expression:

$$V_{\rm H}(x, y, z) = E[{\rm H/surf}](x, y, z) - E[{\rm H_{vac}/surf}]. \quad (1)$$

In Eq. 1, E[H/surf], and $E[{\rm H_{vac}/surf}]$ are the DFT total energies obtained for the H atom placed near the surface in the position x, y, z, and in the middle of the vacuum space between consecutive slabs, respectively. Thus, $E[{\rm H_{vac}/surf}]$ does not depend on the parallel coordinates x, y of the H atom because the interaction with the surface is negligible, and

has been obtained through a spin-polarized calculation as required to properly deal with systems with unpaired electrons. Accordingly, a site-dependent adsorption energy of H on Pt_{1ML}/Ru(0001) can be defined as the minimum value of V_H on each surface site, and the adsorption energy, E_{ads}, is the absolute minimum value of $V_H(x, y, z)$. Note that Eq. 1 entails a negative adsorption energy for stable H chemisorption, and a smaller (more negative) value correspond to a stronger bond.

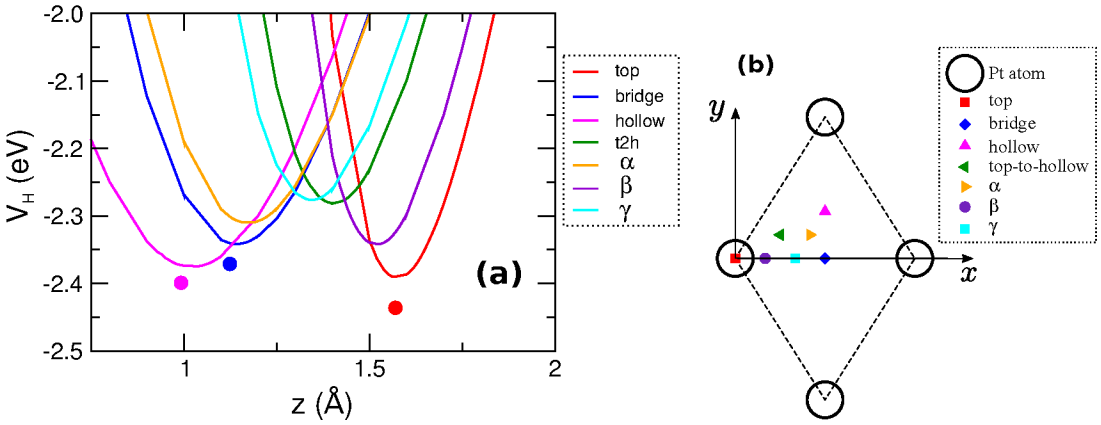


Figure 1: (a) H/Pt_{1ML}/Ru(0001) PES as a function of the distance with respect to surface, z, on various sites. (b) Surface sites considered in panel (a).

Figure 1 (a) shows the z-dependence of the H/Pt_{1ML}/Ru(0001) PES on various surface sites which are schematically shown in Figure 1b. DFT calculations have been first carried out by keeping the surface atoms fixed in their equilibrium positions obtained when the H atom is far form the surface: i.e. rigid surface results (full lines). Then, for the three high symmetry sites for which adsorption is energetically most favorable (*top*, *bridge*, and *hollow-fcc*), we have also included the results obtained allowing adsorption-induced surface relaxation, represented by full circles.

In table 1 we compare the adsorption energies on the *top*, *bridge* and *hollow-fcc* sites, with the DFT values reported by Hoster *et. al.*[11] In agreement with the latter results (computed for a four-layer slab, and the so-called PW91 exchange-correlation functional[41]), we have obtained in both, rigid and relaxed surface calculations, that the most favorable adsorption site is *top*, followed in stability by *hollow-fcc*, and finally by *bridge*.

Quantitatively, the present adsorption energies on all the sites are slightly smaller (stronger bond) than those reported in ref. 11, discrepancies being however, not larger than ~ 0.1 eV. Interestingly, surface relaxation barely affects the H adsorption energies: between 0.03 eV and 0.05 eV depending on the surface site. This validates the use of the rigid surface approximation to describe dissociative adsorption of H_2 on $Pt_{1ML}/Ru(0001)$ since it is precisely on the dissociation products that surface relaxation is expected to play its largest role along the full dissociation pathway.

Table 1: Adsorption energy of H on $Pt_{1ML}/Ru(0001)$ computed without and with surface relaxation (see text) on high symmetry surface sites.

	E_{ads} (eV)		
	top	*bridge*	*hollow-fcc*
Rigid surface	-2.39	-2.34	-2.37
Relaxed surface	-2.44	-2.37	-2.40
Hoster *et. al.*[11]	-2.34	-2.27	-2.28

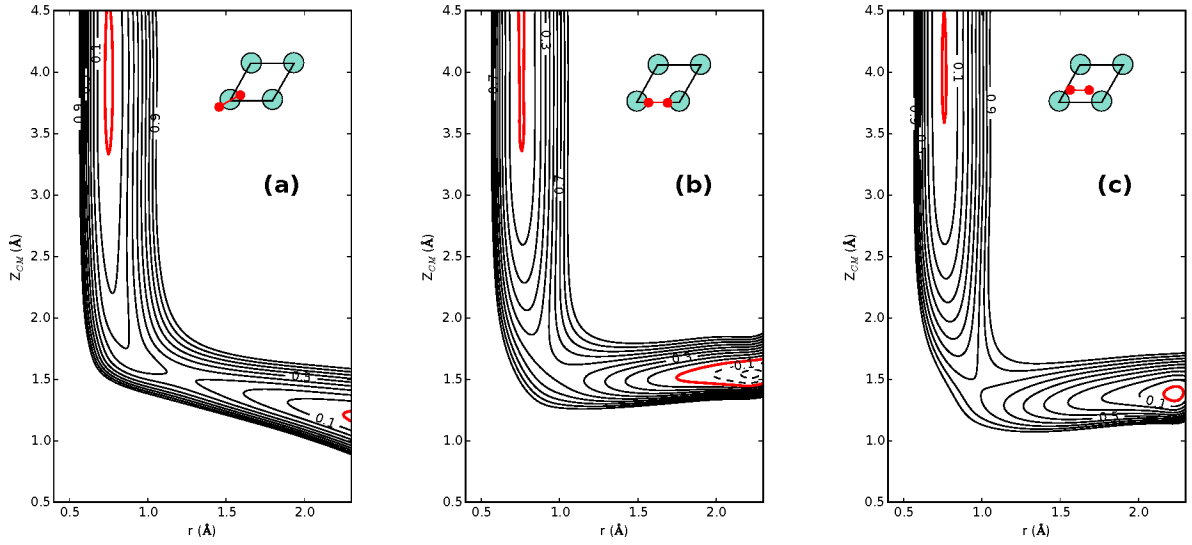


Figure 2: 2D-cuts(r,Z_{cm}) of the $H_2/Pt_{1ML}/Ru(0001)$ PES for the molecular configurations parallel to the surface with the center of mass on *top* (a), *bridge* (b), and *hollow-fcc* (c) sites (the molecular orientations are indicated in the insets). The energy difference between consecutive equipotential lines is $\Delta E = 0.1$ eV, the red lines represent the 0 eV reference energy level corresponding to H_2 and the surface in equilibrium and far from each other, whereas solid (dashed) lines correspond to positive (negative) potential energy values.

Figure 2 shows three 2D-cuts(r,Z_{cm}) of the $H_2/Pt_{1ML}/Ru(0001)$ PES (usually called *elbow*

plots) for the molecule parallel to the surface ($\theta = \pi/2$), and with its center of mass on *top* (a), *bridge* (b), and *hollow-fcc* (c) sites. The molecular azimuthal angles are: in panel (a), $\phi = \pi/6$ (such on-*top*-configuration will be hereafter referred to as *fcc-top-hcp*), and in panels (b) and (c), $\phi = 0$ (see the insets). In Figure 2 (as well as in all the other elbow plots shown in this work), $Z_{\rm cm}$=0 corresponds to the height of metal atoms in the outermost surface layer. The 2D-cuts(r,$Z_{\rm cm}$) for H_2 parallel to the surface and with its center of mass on *top* sites (irrespective the azimuthal angle ϕ) are the ones with the lowest activation energy barrier: E_b=0.32 eV. Thus, a single pseudomorphic monolayer of Pt deposited on Ru(0001) resembles pure Pt(111) in the surface site (*top*) on which the minimum activation energy barrier for dissociation is found.[29] For the molecular configurations considered in panel b (with the center of mass on *bridge*), and (c) (with the center of mass on *hollow-fcc*), the activation energy barriers are 0.45 eV and 0.71 eV respectively. Thus, on both Pt(111) and Pt_{1ML}/Ru(0001), the activation energy barrier tend to increase for configurations with an increasing distance of the molecular center of mass with respect to *top* sites. The main difference between Pt_{1ML}/Ru(0001) and Pt(111) is certainly the value of E_b: 0.32 eV vs. 0.06 eV.[29]

It is interesting to compare the PES of H_2/Pt_{1ML}/Ru(0001) with that of H_2/Pd_{1ML}/Ru(0001).[42,43] The activation energy barrier found for the *fcc-top-hcp* configuration for both, H_2/Pt_{1ML}/Ru(0001) and H_2/Pd_{1ML}/Ru(0001) are very close to each other (0.32 eV vs. 0.34 eV). Still, in the latter case the molecules can reach $Z_{\rm cm} \sim 1.7$ Å without experiencing any repulsive interaction with the surface.[42] In contrast, the lowest value of the potential energy for such a small $Z_{\rm cm}$ value for H_2 in a *fcc-top-hcp* configuration on Pt_{1ML}/Ru(0001) is 0.2 eV. Thus, the main difference between the *fcc-top-hcp* 2D-cuts(r,$Z_{\rm cm}$) of the H_2/Pd_{1ML}/Ru(0001) and H_2/Pt_{1ML}/Ru(0001) PESs are found in the entrance channel: in the former case the potential is flat and slightly attractive up to $Z_{\rm cm} \sim 1.7$ eV, whereas in the latter is repulsive already relatively far from the surface (see Figure 2a and the left panel of Figure 3a of ref. 42). This slightly attractive character of the H_2/Pd_{1ML}/Ru(0001) PES in the entrance channel provokes some trapping

of very low energy impinging H_2 molecules that favors the probability for a molecule to find an active site like an isolated Ru atom in the case of a Pd-rich Pd_xRu_{1-x}/Ru(0001) surface alloy.[19] Such an effect is expected to play a less important role in the case of Pt-rich Pt_xRu_{1-x}/Ru(0001) surface alloy due to the absence of long distance molecule-surface attraction.

Using the H_2/Pt_{1ML}/Ru(0001) interpolated PES, we have computed the initial dissociative adsorption probability, P_{diss}, through QCT calculations. In Figure 3a we plot P_{diss} as a function of the translational kinetic energy perpendicular to the surface, $E_\perp = E_i\ cos^2\theta_i$, for H_2 molecules initially in its ground ro-vibrational state ($\nu = 0$,J=0) for various angles of incidence, $0 \leq \theta_i \leq 60^\circ$ (θ_i=0 corresponds to normal incidence). The rapid increase of $P_{diss}(E_\perp)$ for $E_\perp \gtrsim 0.15$ eV (for small incidence angles) is a direct consequence of the activated character of dissociative adsorption. The fact that P_{diss} is greater than zero for impact energies lower than the lowest minimum activation energy barrier mentioned above (i.e. 0.32 eV) is simply due to the energy flow from the vibrational coordinate (i.e. the vibrational zero point energy, ZPE) to the reaction coordinate. If the ZPE is not taken into account (i.e. Classical calculations) and for normal incidence (dashed line in Figure 3a), $P_{diss}(E_i)$ presents a threshold very close to the minimum activation energy barrier.

Concerning the dependence with the angle of incidence, the results for $\theta_i \leq 40^\circ$ for any given value of $E_\perp$, agree well with each other. In this range of angles of incidence, P_{diss} scales with normal energy (normal energy scaling) as often observed for activated dissociative adsorption of H_2 on other metal surfaces. However, for θ_i=60° the $P_{diss}(E_\perp)$ curve is well below all the ones for smaller θ_i values. For a given value of $E_\perp$, for $\theta_i \gtrsim 45^\circ$, larger values of parallel initial translational kinetic energy entail smaller values of P_{diss}. This is likely due to a shadowing effect[44] provoked by sites where the PES is more repulsive in the entrance channel (e.g. *hollow*) which prevents molecules impinging under near grazing incidence conditions, to approach the surface and encounter the lowest barriers on the most reactive sites (i.e. *top*). Figure 3b shows the distribution of closest approach distance to the surface, Z_{min}, for

reflected trajectories with $E_\perp$=0.3 eV and θ_i=0°, 40°, and 60°. As expected, and in line with a shadowing effect mentioned above, larger incidence angles entail reflection further away from the surface in spite of the same $E_\perp$ value (0.3 eV) in the three cases.

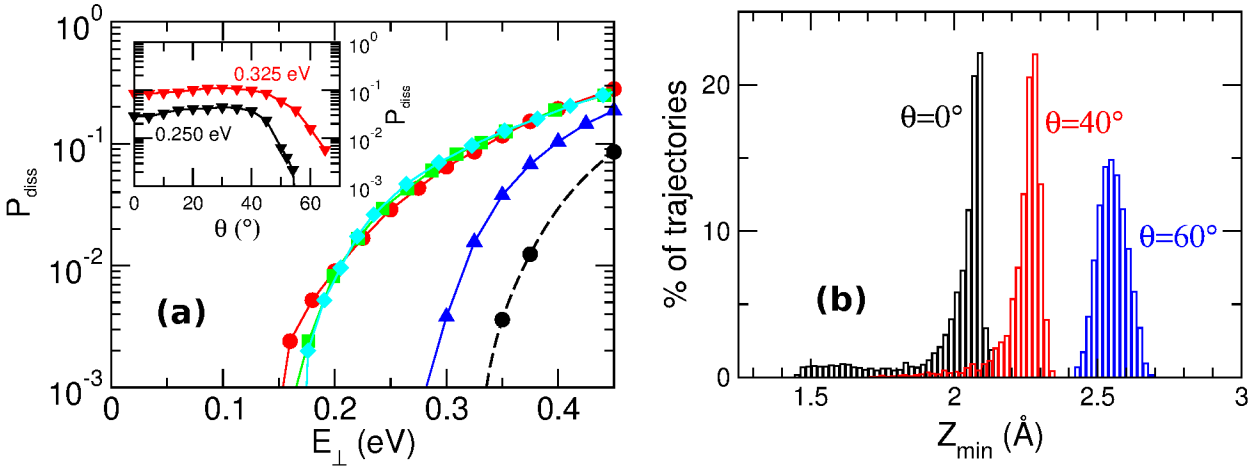


Figure 3: (a) Quasi-classical $P_{diss}(E_\perp)$ cuurve for $H_2(\nu$=0,J=0)/Pt_{1ML}/Ru(0001) for: $\theta_i = 0°$ (red line with circles), $\theta_i = 20°$ (green line with squares), $\theta_i = 40°$ (cyan line with diamonds) and $\theta_i = 60°$ (blue line with triangles). Dashed black line with circles: Classical P_{diss} for $\theta_i = 0°$. Inset of panel (a): $P_{diss}(\theta_i)$ for $E_\perp$=0.25 eV (black line with symbols), and 0.325 eV (red line with symbols). (b) Distribution of closest approach distance to the surface, Z_{min}, for reflected trajectories with $E_\perp$=0.3 eV for: θ_i=0° (black), θ_i=40° (red), θ_i=60° (blue).

Since in general, a non negligible fraction of molecules impinging a surface are ro-vibrationally excited, we have computed P_{diss} for various molecular initial states $H_2(\nu,J)$. In Figure 4a we have plotted P_{diss} as a function of $E_\perp$ for $H_2(\nu,J)$ with $(\nu = 0, J = 0, 2, 4, 8)$ and $(\nu = 1, J = 0)$, always for normal incidence (θ_i=0). For impact energies larger than $\sim$ 0.25 eV, for which $P_{diss}(\nu$=0,J=0) $\gtrsim$ 0.04, P_{diss} is barely affected by the initial rotational quantum number J (except for quite high J values, e.g. J=8). However, for lower values of $E_\perp$, rotational excitation promotes dissociation. This is not surprising since near the dissociation threshold for which P_{diss} is very small, initial rotational excitation turns energetically accessible a non-negligible fraction of dissociation pathways. Thus, the threshold of $P_{diss}(E_\perp)$ shifts down when J increases. Initial vibrational excitation has a more pronounced effect on reactivity. For instance, for $H_2(\nu$=1,J=0) P_{diss} is more than two orders of magnitude

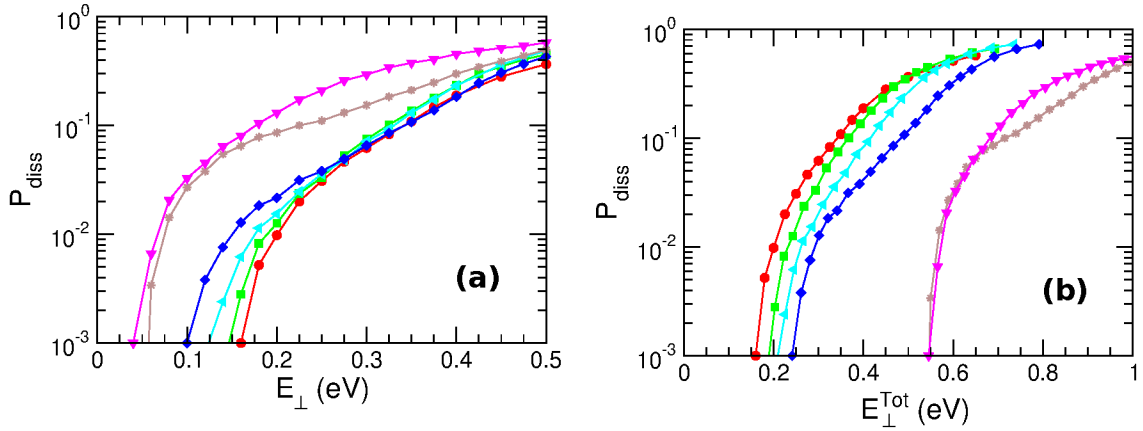


Figure 4: $P_{diss}(E_\perp)$ (a), and $P_{diss}(E_\perp^{tot})$ (b) for $H_2(\nu,J)/Pt_{1ML}/Ru(0001)$ under normal incidence (see text). ν=0,J=0 (red circles), ν=0,J=2 (green squares), ν=0,J=3 (cyan triangles), ν=0,J=4 (blue diamonds), ν=0,J=8 (brown stars), ν=1,J=0 (magenta triangles).

larger than for $H_2(\nu$=0,J=0) at the same value of $E_\perp < 0.15$ eV, and even well above the threshold for ν=0,J=0 (e.g. for $E_\perp$=0.4 eV), $P_{diss}(\nu$=1,J=0) is larger than $P_{diss}(\nu$=0,J=0) by more than a factor of 2. However, it must be considered that, except for large J values, vibrationally excited states entail a larger internal energy of the molecule than rotationally excited states. For instance, the energy gap between $H_2(\nu$=1,J=0) and $H_2(\nu$=0,J=0) is $\sim$ 0.51 eV, whereas the one between $H_2(\nu$=0,J=2) and $H_2(\nu$=0,J=0) is only $\sim$ 0.04 eV, and a J-value as large as 8 is required for a rotationally excited state $H_2(\nu$=0,J) to reach an internal energy of the molecule close to that of $H_2(\nu$=1,J=0). In order to compare the efficacy of initial vibrational and rotational energy (to promote dissociative adsorption), in Figure 4b we have plotted the results of Figure 4a but as a function of $E_\perp^{tot}$:

$$E_\perp^{tot} = E_\perp + E_{int}[H_2(\nu, J)] - E_{int}[H_2(\nu = 0, J = 0)], \quad (2)$$

being $E_{int}[H_2(\nu$,J)] the internal energy of H_2 in the ro-vibrational state (ν,J) in vacuum; i.e. $E_{int}[H_2(\nu$=0,J=0)]=ZPE. When plotted as a function of $E_\perp^{tot}$, the sticking curves for all the initial excited states are below the one for $H_2(\nu$=0,J=0), which shows that translation

perpendicular to the surface is the degree of freedom (DOF) in which energy is most efficient to promote dissociation. In other words, rotational and vibrational efficacy[45] is smaller than one. Interestingly, the sticking curves of $H_2(\nu{=}0,J{=}8)$ and $H_2(\nu{=}1,J{=}0)$ in Figure 4b are very close to each other which shows that vibrational and rotational motion are almost equally efficient promoting dissociation, being the total internal energy of the molecule which really matters.

To further characterize the mechanism of H_2 dissociative adsorption on $Pt_{1ML}/Ru(0001)$, we have evaluated the mean number of rebounds, $\langle N_{reb} \rangle$, of trajectories before dissociation and reflection. For $H_2(\nu{=}0,J{=}0)$, normal incidence, and $E_\perp{=}0.2$ eV, we have found $\langle N_{reb} \rangle$ $=0.96$ for trajectories ended in dissociative adsorption and $\langle N_{reb} \rangle$ $=1.12$ for reflected ones, which shows signs of trapping neither in the reactive nor in the unreactive channel. This is in line with the lack of any attractive character of the $H_2/Pt_{1ML}/Ru(0001)$ PES in the entrance channel as mentioned above. Since such an indirect mechanism becomes less relevant when $E_\perp$ increases, and it is not expected to be significantly influenced by the initial ro-vibrational state of the molecules, similar conclusions are assumed valid for other initial conditions above the dissociative adsorption threshold.

Figure 5a shows that for $E_\perp{=}0.2$ eV, dissociation only takes place near *top* sites. Moreover, for normal incidence, only molecules starting near *top* sites dissociate. This is due to the fact that the lowest activation energy barrier is located on *top* sites, and also shows that neither trapping nor steering play a significant role. Otherwise, molecules initially placed on sites characterized on high activation energy barriers might finally encounter an energetically low energy pathway after exploring the surface for a while or being directly steered by the molecule-surface interaction potential. In line with this, Figure 5b shows that something similar occurs with the polar angle distribution of reactive molecules: only initially well oriented molecules (i.e. parallel to the surface, $70^\circ \leq \theta_i \leq 115^\circ$) finally dissociate also in a nearly flat configuration.

Before concluding the analysis of the reactivity of $Pt_{1ML}/Ru(0001)$, and in order to il-

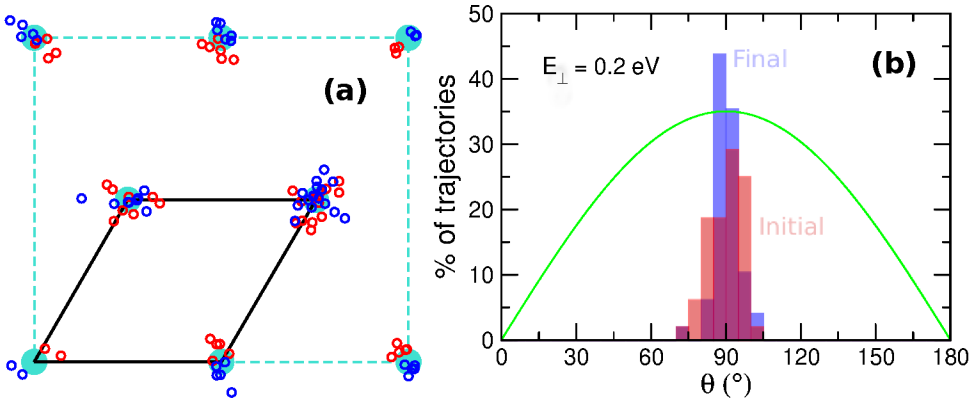


Figure 5: (a) Initial (open red circles) and final (open blue circles) positions of the molecular center of mass of reactive trajectories for $E_\perp$=0.2 eV. Cyan circles represent the topmost layer Pt atoms, and the black lines indicate the unit cell. (b) Initial (red bars) and final (blue bars) distribution of the polar angle θ of reactive trajectories for $E_\perp$=0.2 eV. The green line represents the $\sin\theta$ function corresponding to the initial random distribution of molecular orientations.

lustrate the effect of alloying, in Figure 6 we compare with the sticking probabilities of $H_2(\nu{=}0,J{=}0)$ on Pt(111)[30] and Ru(0001)[31,46] (all at normal incidence). Pt_{1ML}/Ru(0001) is less reactive than both Pt(111), and Ru(0001), which illustrates the versatility of bimetallic surfaces whose properties are not necessarily intermediate between the ones of the constituent pure metals. In this case, strain and ligand effects make Pt_{1ML}/Ru(0001) significantly less reactive than Pt(111) which in turn, is less reactive than Ru(0001). The H_2 dissociation threshold on Pt_{1ML}/Ru(0001) is higher than on Pt(111) as expected in view of the larger activation energy barrier in the former case. However, the difference between both dissociation thresholds is significantly smaller than $E_b^{Pt_{1ML}/Ru(0001)}$-$E_b^{Pt(111)}$=0.26 eV, and also the shape of the $P_{diss}(E_i)$ curves for both systems differs significantly with respect to each other. To illustrate this, in Figure 5 we have also plotted the $P_{diss}(E_i)$ curve of $H_2(\nu{=}0,J{=}0)$/Pt(111) horizontally shifted to the right by 0.26 eV (dashed line). Though for $E_i \gtrsim 0.35$ eV the latter curve is quite close to the sticking curve for H_2/Pt_{1ML}/Ru(0001), at lower energies the reactivity of Pt_{1ML}/Ru(0001) is much higher than one might predict through such a simple shifting procedure. This is because for H_2/Pt_{1ML}/Ru(0001), the dissociation threshold ($\sim$ 0.15 eV) is significantly lower than the minimum activation energy barrier for dissocia-

tion (E_b=0.32 eV), in contrast with H_2/Pt(111) (and also with H_2/Ru(0001)). As already mentioned, this is due to the initial ZPE of H_2 which in the case of Pt_{1ML}Ru(0001), enhances dissociative adsorption more efficiently than for Pt(111) and Ru(0001) due to the later character of the activation energy barriers.

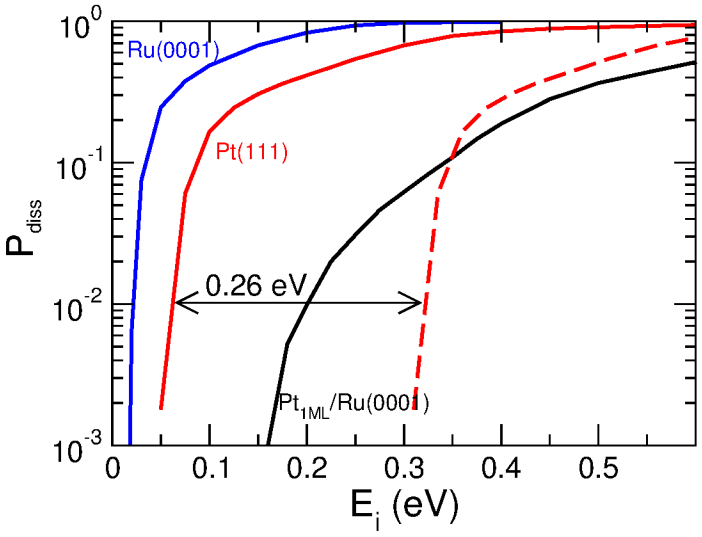


Figure 6: $P_{diss}(E_i)$ of $H_2(\nu$=0,J=0) on Pt_{1ML}/Ru(0001) (black line, present work), Pt(111) (red line, taken from ref. 30), and Ru(0001) (blue line, taken from ref. 46) for normal incidence. Dashed red line: $P_{diss}(E_i)$ curve for $H_2(\nu$=0,J=0)/Pt(111), shifted to the right by 0.26 eV (see text).

In spite of this significant vibrationally efficacy around the threshold for Pt_{1ML}/Ru(0001), the dissociative adsorption probability of low energy H_2 molecules ($E_i \lesssim 0.1$ eV) impinging a single pseudomorphic Pt overlayer deposited on Ru(0001) is smaller than on Ru(0001) by several orders of magnitude. This reactivity of Pt_{1ML}/Ru(0001) smaller than that of Ru(0001) is in good qualitative agreement with experiments.[17,18] However, the measured initial sticking probabilities of D_2 on a Pt_xRu_{1-x}/Ru(0001) surface alloy with $\sim$ 100 % Pt,[17] and on Pt islands deposited on Ru(0001) for Pt coverage of 1.2-1.4 ML[18] is only $\sim$ 30 % and $\sim$ 65 % smaller than on Ru(0001) respectively (under the same experimental conditions). To try to understand this apparent quantitative discrepancy between theory and experiments, one should keep in mind that on the one hand, in the case of Pt islands deposited at room temperature on Ru(0001), the lack of annealing at high enough temperatures, prevents the

formation of a Pt_xRu_{1-x} alloy and for instance, for 1.2 ML, the surface exposes 25 % of a two-layer Pt film, 66 % of a single Pt layer, and also 9 % of substrate Ru atoms due to *holes* in the Pt monolayer.[13] Thus, the relatively high experimental reactivity (only 65 % smaller than that of Ru(0001)),[18] might come from two-monolayer thick Pt islands, exposed underlying Ru atoms, and/or undercoordinated Pt atoms in the borders between islands of different thickness. To investigate this, in subsection 3.2 we consider the reactivity of a pseudomorphic Pt bilayer and the effect of Pt-*holes* in both, Pt_{1ML}/Ru(0001) and Pt_{2ML}/Ru(0001). On the other hand, in the case of the surface alloys Pt_xRu_{1-x} (prepared after annealing above $\sim$ 1100 K of the Pt overlayer), some *isolated* Ru atoms might still be exposed even for a Pt exposure $\sim$ 1 ML. Therefore, in subsection 3.3 we will consider also the reactivity of isolated Ru monomers and dimers in a Pt-rich Pt_xRu_{1-x} surface alloy.

3.2 Full Pt multilayers, and Pt overlayers with vacancies on Ru(0001)

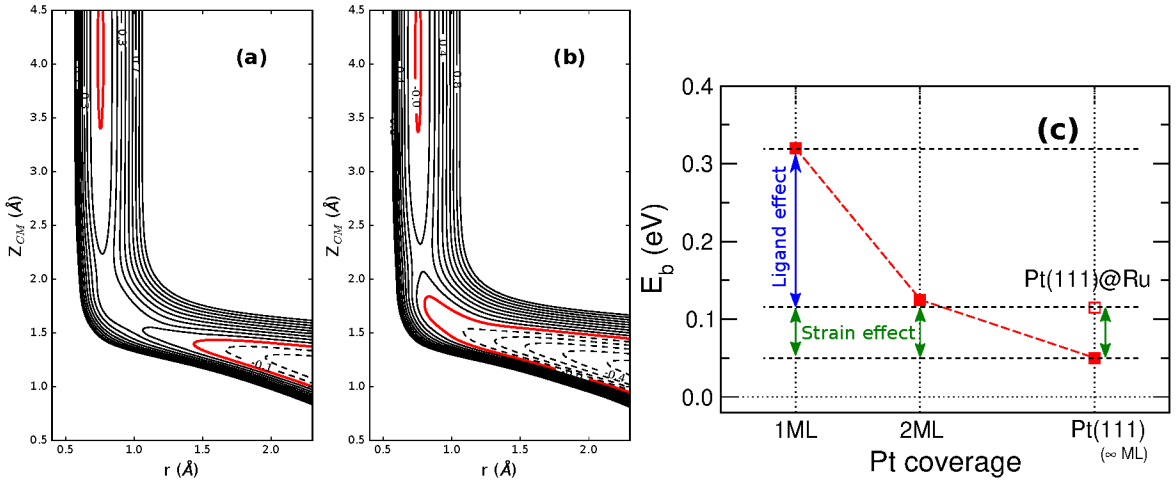


Figure 7: 2D-cuts(r,Z_{cm}) for the *fcc-top-hcp* configuration for H_2 on Pt_{2ML}/Ru(0001) (a), and Pt(111)@Ru (b) (see text). (c) Activation energy barrier extracted from the *fcc-top-hcp* 2D-cuts(r,Z_{cm}) for H_2 on Pt_{1ML}/Ru(0001), Pt_{2ML}/Ru(0001), Pt(111)@Ru (present work), and Pt(111) (taken from ref. 31.)

In Figure 7a we show a 2D-cut(r,Z_{cm}) of the H_2/Pt_{2ML}/Ru(0001) PES for a molecule parallel to the surface with its center of mass on a *top* site and the H-H bond pointing to *hollow* sites, i.e. the most favorable configuration for H_2 dissociation on Pt_{1ML}/Ru(0001)

and Pt(111). In this case, the activation energy barrier is 0.13 eV: 0.19 eV smaller than for a similar molecular configuration on Pt_{1ML}/Ru(0001) (Figure 2a). Interestingly, the activation energy barrier obtained for the same molecular configuration on Pt(111)@Ru (Figure 7b) differs by only 0.01 eV from the one on Pt_{2ML}/Ru(0001). This means that for two pseudomorphic Pt overlayers on Ru(0001), vertical ligand effects barely affects the height of the activation barrier. Thus, it is expected that a further increase of the thickness of the pseudomorphic Pt film will not modify the size of the activation energy barrier for H_2 dissociative adsorption. Still, comparing the 2D-cuts(r,Z_{cm}) for H_2/Pt_{2ML}/Ru(0001) (Figure 7a) and H_2/Pt(111)@Ru (Figure 7b) it is observed that the activation barrier in the latter case is displaced towards the entrance channel in a position very similar to the one for pure Pt(111).[29] Thus, for H_2/Pt_{2ML}/Ru(0001) ligand effects barely affect the barrier height but still play some role in its location. Comparing the minimum activation energy barriers for H_2 dissociation on Pt(111)@Ru and Pt(111), it is possible to isolate and quantify the role of strain and ligand effects for H_2/Pt_{1ML}/Ru(0001) and H_2/Pt_{2ML}/Ru(0001). Taking always H_2/Pt(111) as a reference, strain effects are the responsible for a 0.06 eV increase of the activation energy barrier for H_2 dissociation on psudomorphic Pt overlayers on Ru(0001) (irrespective the number of Pt layers). Then, the exceeding 0.2 eV increase found in the case of H_2/Pt_{1ML}/Ru(0001) can be ascribed to ligand effects only. This is schematically illustrated in Figure 7c. The relative role of strain and ligand effects as a function of the number of Pt overlayers, on the activation energy barrier is similar to that found by Hoster and co-workers for the adsorption energy of H atoms.[11]

Far from the borders of pseudomorphic Pt islands the activation energy barriers for H_2 dissociation are expected to be similar to the ones obtained for the full Pt overlayers of the same thickness. In contrast, near the borders the local reactivity is expected to be influenced by the presence of: i) undercoordinated atoms in the step-edge, and ii) underlying Ru or Pt atoms exposed between islands or in *holes* of the overlayers. In order to model these situations, we have generated an aggregate of three Pt-atom vacancies in Pt_{1ML}/Ru(0001).

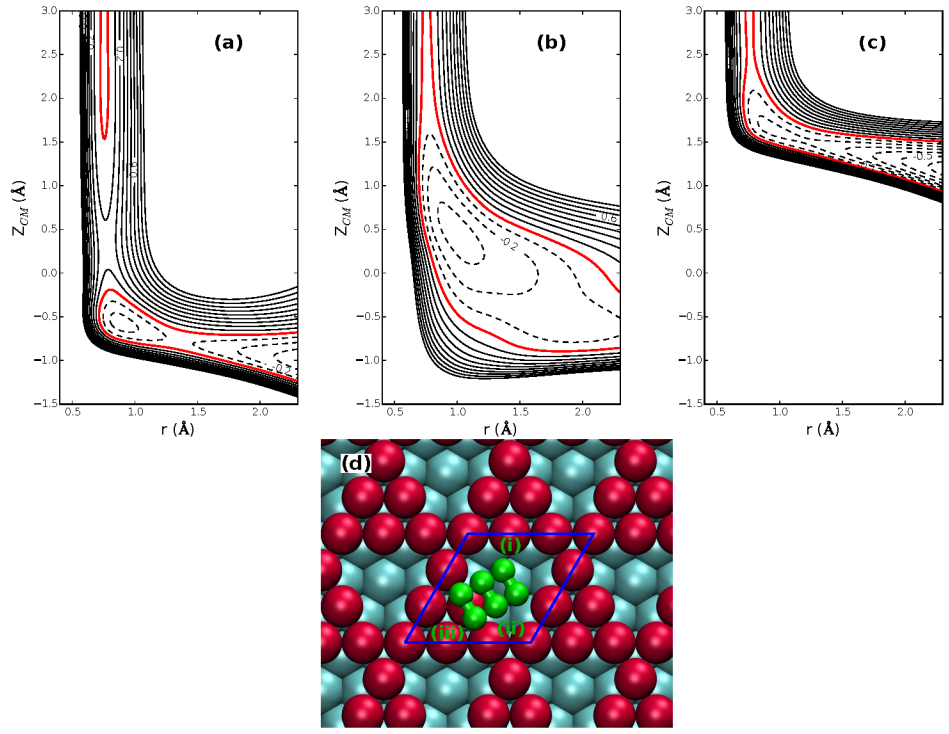


Figure 8: (a), (b) and (c): 2D-cuts(r,Z_{cm}) for H_2 on Pt_{1ML}/Ru(0001) with three Pt vacancies in the topmost layer. (d) Schematic diagram of the molecular configurations (i), (ii), and (iii) considered in panels (a), (b) and (c) respectively. Red (cyan) spheres represent Pt (Ru) atoms.

Then, we have computed the H_2/surface PES along 2D-cuts(r,Z_{cm}) for the molecule parallel to the surface with its center of mass on various positions along the Pt-*hole*. In Figure 8a we show the results for the molecule on top of the exposed Ru atom in the center of the Pt-*hole* where the activation energy barrier for H_2 dissociation is 0.14 eV, i.e. $\sim$ 0.12 eV higher than for a similar configuration on Ru(0001). This reduction of reactivity with respect to Ru(0001) can be attributed only to ligand effects since the underlying Ru layers are not under strain/stress. Pt-*holes* as well as the borders of pseudomorphic Pt islands entail the existence of undercoordinated Pt atoms which in addition, are expected to be less affected by strain than those in a full overlayer, both effects being expected to increase reactivity. In Figs. 8c and b we have plotted 2D-cuts(r,Z_{cm}) of the PES for H_2 with its center of mass on top of a Pt atom in the border of the Pt-*hole* and midway towards the center of the hole respectively (see Figure 8d). In both cases dissociative adsorption is a non activated process. Thus, holes in a full Pt monolayer deposited on Ru(0001) are expected to produce a significant increase of the local reactivity of the bimetallic surface, being the main responsible for that, the undercoordinated Pt atoms in the borders of the hole. Similar results are also expected in the case of monoatomic Pt islands in the Pt-sub-monolayer regime, being the Pt atoms in the edges of the island, the most active sites for H_2 dissociation.

In view of these results, non-activated dissociation pathways are also expected near the border of an aggregate of three Pt vacancies in the topmost layer of Pt_{2ML}/Ru(0001). This is in fact the case as illustrated in Figure 9. Whereas in the center of the hole (i.e. on top of a lowest-layer Pt atom exposed) there is an activation energy barrier of 0.31 eV for H_2 dissociation (Figure 9 (a)), on top of a topmost-layer Pt atom in the border of the hole, dissociation is non activated (Figure 9 (c)). In an intermediate configuration (Figure 9 (b)), there is a potential well similar but deeper than the one observed in (Figure 8 (b)). Interestingly, a low energy molecule impinging parallel to the surface in a wide fraction of a three Pt-hole in both, Pt_{1ML}/Ru(0001), and Pt_{2ML}/Ru(0001), can approach the surface up to $Z_{cm} \sim$ 1.5-2.0 Å without encountering an activation energy barrier. Even molecules

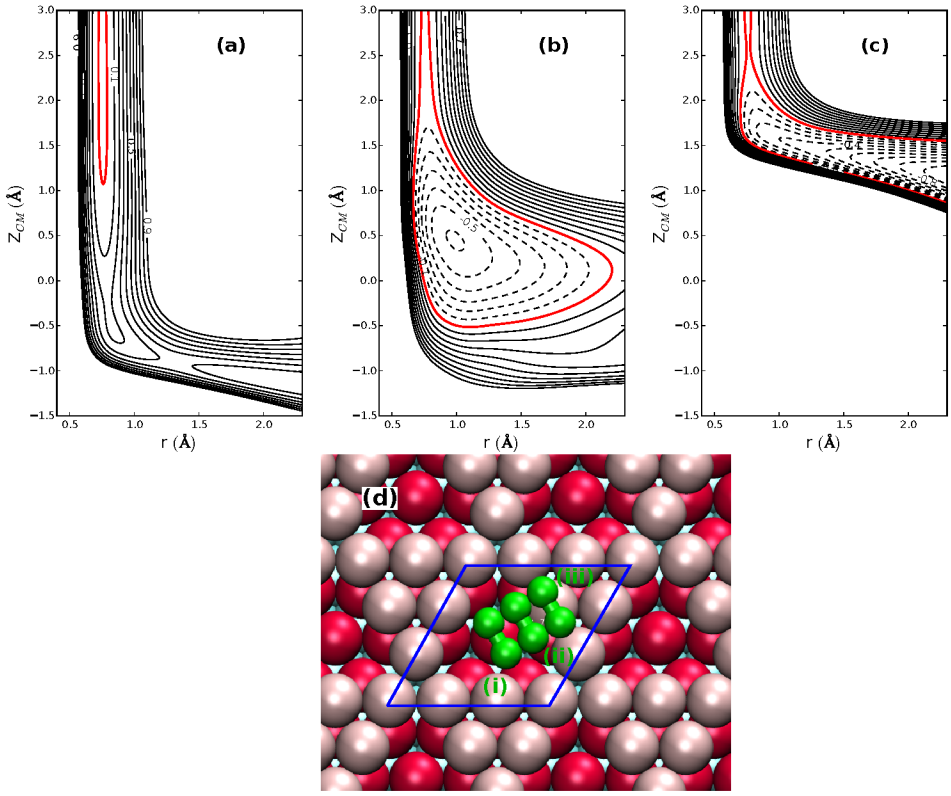


Figure 9: Idem Figure 8 for H_2 on Pt_{2ML}/Ru(0001) with three Pt vacancies in the topmost layer. In panel (d), pink (red) spheres represent Pt atoms in the topmost (second) surface layer.

impinging the center of the hole (the less reactive region) still can move laterally towards the borders and dissociate without encountering an activation energy barrier at all. Thus, the present results suggest that the initial sticking probability of Pt island deposited on Ru(0001) for a Pt coverage $\sim$ 1.2-1.4 ML ($P_{diss} \sim 0.04$)[18] might come mainly from undercoordinated Pt atoms in the borders of Pt holes (which represent the $\sim$ 9 % of the bimetallic surface for 1.2 ML of Pt[13]) in one-Pt-layer areas, and in the borders of two Pt-layer thick islands on which dissociative adsorption is non activated. Some contribution might also come from dissociation near the center of two Pt-layer thick islands which cover $\sim$ 25 % of the bimetallic surface and where the activation energy barrier for H_2 dissociation is $\sim$ 0.2 eV smaller than on Pt_{1ML}/Ru(0001). Unfortunately, to evaluate in what extent direct steering by the potential, and/or if an indirect dynamic trapping mediated mechanism might enhance the cross section of dissociation on undercoordinated Pt atoms for low energy H_2 molecules, dynamics calculations are required and this is beyond the scope of the present work.

3.3 Pt-rich Pt_xRu_{1-x}/Ru(0001) surface alloys

The experimental initial sticking probability of H_2 on a Pt_xRu_{1-x}/Ru(0001) surface alloy with $\sim$ 100 % of Pt ($x \sim 1$), only a factor of two smaller than on Ru(0001)[17] cannot be explained by the results of subsection 3.1 which predict a difference of several orders of magnitude. To explore possible reasons for such apparent quantitative discrepancy we have explored the reactivity of isolated Ru and Ru_2 aggregates in a Pt-rich Pt_xRu_{1-x}/Ru(0001) surface alloy. Thus, in Figs. 10 and 11 we present various 2D-cuts(r,Z_{cm}) of the PES of H_2/Pt_xRu_{1-x}/Ru(0001) with x=0.89 and x=0.78 respectively. In all the cases we have considered molecular configurations parallel to the surface with the center of mass on top, and bridge sites, the first and second most reactive high-symmetry sites for H_2 dissociation on Pt_{1ML}/Ru(0001) (present work), Pt(111),[29] and Ru(0001).[31] For both Pt concentrations, H_2 dissociation on top of a Ru atom is a non activated process. Thus, in line with previous results for Pd_xRu_{1-x}/Ru(0001),[19] isolated Ru atoms in the surface alloy are more reactive

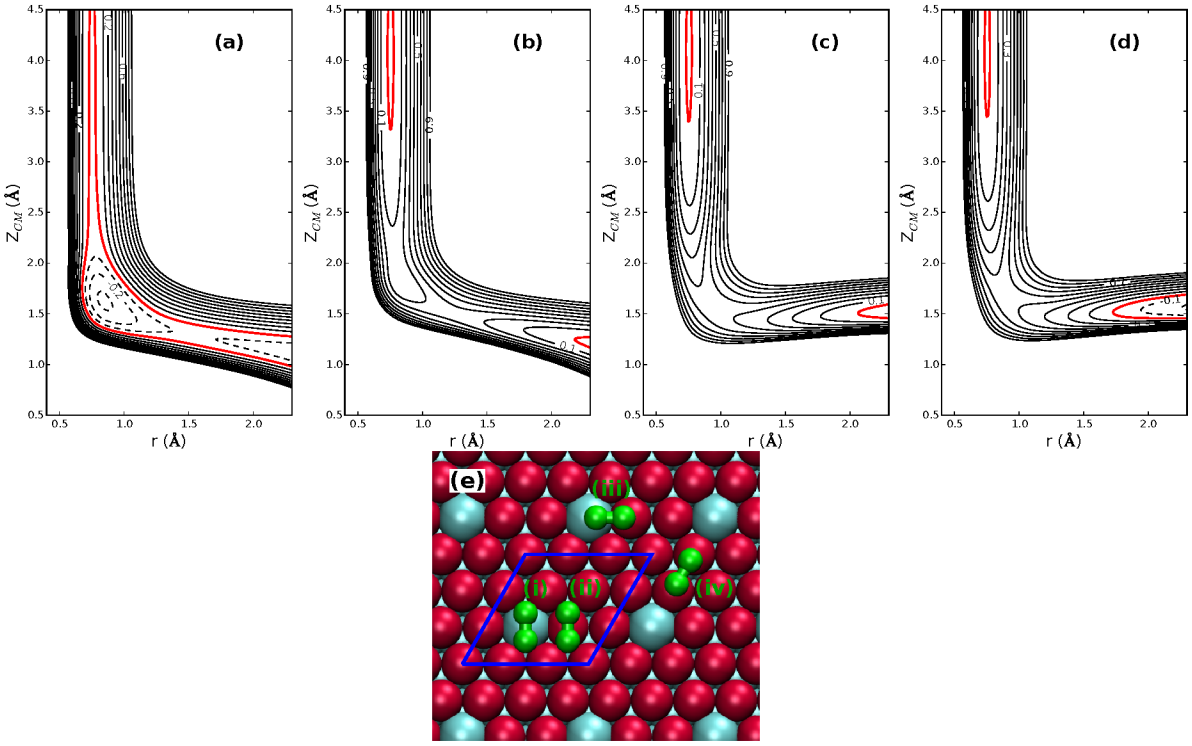


Figure 10: (a), (b), (c) and (d): 2D-cuts(r,Z_{cm}) for $H_2/Pt_{0.89}Ru_{0.11}/Ru(0001)$. (e) Schematic diagram of the molecular configurations (i), (ii), (iii), and (iv) considered in panels (a), (b), (c), and (d) respectively. Red (cyan) spheres represent Pt (Ru) atoms.

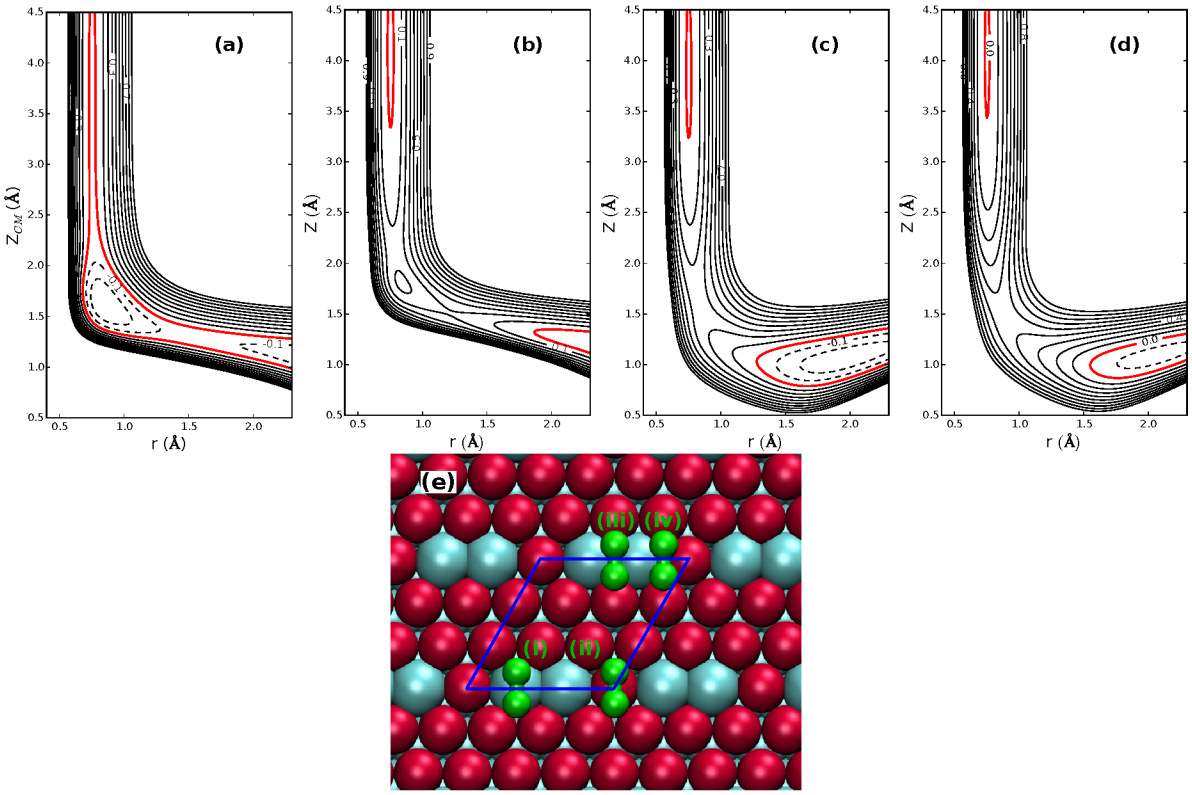


Figure 11: Idem Figure 10 for $H_2/Pt_{0.78}Ru_{0.22}/Ru(0001)$

than in pure Ru(0001). Various Pt atoms surrounding a Ru atom of the topmost layer of the Pt_xRu_{1-x} surface alloy, turn it more reactive which certainly represents a case of enhanced reactivity due to lateral ligand effects. On top of Pt atoms, the activation energy barrier for H_2 dissociation is $\sim$ 0.2 eV for both surface alloys: for x=0.89 (Figure 10 (b)), and for x=0.78 (Figure 11 (b)). This activation energy barrier is $\sim$ 0.1 eV smaller than on a top site of $Pt_{1ML}/Ru(0001)$ which means that isolated Ru atoms in the Pt-rich $Pt_xRu_{1-x}/Ru(0001)$ surface alloy also activate its NN Pt atoms. In any $Pt_xRu_{1-x}/Ru(0001)$ surface alloy there are in principle three types of bridge sites: i) Pt-Pt, ii) Pt-Ru, and iii) Ru-Ru. In our model of the $Pt_{0.89}Ru_{0.11}/Ru(0001)$ surface alloy, there is no type-iii bridge sites since there is only one Ru atom per supercell, and the activation energy barriers on both type-i and type-ii bridge sites are very similar to each other ($\sim$ 0.45 eV, see Figs. 10 (c) and (d)) and also close to that of bridge sites in $Pt_{1ML}/Ru(0001)$ (Figure 2 (b)). This might be due to the fact that the activation energy barrier on bridge sites on both pure surfaces, Ru(0001)[31] and

Pt(111)[29] are also very close to each other ($\sim$ 0.35 eV). In contrast, a significant difference is observed between the activation energy barriers on type-iii and type-ii bridge sites of $Pt_{0.78}Ru_{0.22}$/Ru(0001) considered in Figs. 11 (c) and (d) respectively. Whereas the activation energy barrier in the 2D-cut(r,Z_{cm}) for the type-iii bridge site $E_b \sim 0.25$ eV (Figure 11 (c)), on the type-ii one, $E_b \sim 0.45$ eV (Figure 11 (d)). In view of these results, the global reactivity of a Pt-rich Pt_xRu_{1-x}/Ru(0001) surface alloy is expected to be largely dominated by isolated Ru atoms or small Ru aggregates on which dissociation is non-activated (on top-Ru sites). Thus, according with the present study, the relatively high initial sticking probability observed experimentally for Pt_xRu_{1-x}/Ru(0001) with $x \sim 1$ ($P_{diss} \sim 0.12$) could be due to isolated Ru atoms and/or small Ru aggregates, or defects (e.g. steps) in the almost full Pt monolayer deposited on Ru(0001). Still, it must be emphasized that here we consider only the initial sticking probability of H_2; i.e. on *clean* PtRu bimetallic surfaces. Since adsorbed H atoms should bound to the active sites for H_2 dissociation (e.g. isolated Ru, and undercoordinated Pt atoms) more strongly than on *perfect* Pt overlayers, the H diffusion out of these sites (i.e. spillover) certainly plays a key role for the surface reactivity in presence of pre-adsorbed H atoms.[47] However, the study such H-coverage effects is beyond the scope of the present work.

4 Conclusions

In this work we present a Density Functional Theory (DFT) study of H_2 dissociative adsorption on various bimetallic PtRu surfaces that can be produced by evaporating Pt on Ru(0001) followed by annealing. Depending on the annealing temperature and Pt coverage, Pt atoms form pseudomorphic islands of variable thickness (for annealing temperature $\sim$ 650 K), or a single layer Pt_xRu_{1-x} surface alloy (for annealing temperature $\sim$ 1300 K and a Pt coverage up to $\sim$ 1 ML). Therefore, we have considered the reactivity of one- and two-Pt pseudomorphic monolayers (including the effect of Pt vacancies), and Pt-rich Pt_xRu_{1-x} surface alloys

($x \lesssim 1$). In the case of a single Pt monolayer, Pt_{1ML}/Ru(0001), we have also computed the sticking probability of H_2 under various impact conditions through classical trajectory calculations by using a continuous representation of the PES obtained by interpolation of the DFT data. We have found that H_2 dissociative adsorption on Pt_{1ML}/Ru(0001) presents a minimum activation energy barrier of 0.32 eV, i.e. larger than on both pure Pt(111) and Ru(0001) by ~ 0.26 eV and ~ 0.3 eV respectively. Compared with pure Pt(111), this lower reactivity of Pt_{1ML}/Ru(0001) is due to vertical ligand effects and also (in less extent) to strain due to the lattice mismatch between Pt and Ru. This entails a sticking probability for low energy H_2 molecules ($E_i \lesssim 0.1$ eV) impinging on Pt_{1ML}/Ru(0001), several orders of magnitude smaller than for Ru(0001). This is not consistent with the experimental initial sticking probability of D_2 found for PtRu bimetallic surface alloys produced by evaporation of $\sim$ 1.0-1.2 ML of Pt on Ru(0001) followed by annealing at 1300 K (Pt_xRu_{1-x} surface alloy) and 650 K (pseudomorphic Pt islands), which are only 30 % and 65 % smaller than on Ru(0001) respectively. In view of our results, such relatively large initial sticking probabilities of low energy hydrogen molecules might be due to undercoordinated Pt atoms in the latter case, or to the existence of isolated Ru atoms or small Ru aggregates in the former.

Acknowledgement

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Graphical TOC Entry

